

# GLRT-Based Coherent Detection in Sub-Gaussian Symmetric Alpha-Stable Clutter

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**Abstract**—This letter presents a generalized likelihood ratio test (GLRT)-based adaptive detector specially for the sub-Gaussian symmetric alpha-stable (SGS $\alpha$ S) sea clutter background. Since the probability density function (PDF) of SGS $\alpha$ S distribution cannot be expressed as a closed-form expression in terms of the elementary function, the research on target detectors in alpha-stable clutter background is very limited. In this letter, firstly, the Fox's H-function is adopted to formulate the PDF of the SGS $\alpha$ S distribution so that the PDF of SGS $\alpha$ S distribution can be expressed as a closed form in terms of H-function. Then, the GLRT-based detector in alpha-stable clutter is designed based on the two-step GLRT criterion and the closed-form expression of the test statistics in terms of the H-function is given. Experimental results based on the simulated data and the measured data verify the effectiveness of the proposed detector.

**Index Terms**—Alpha-stable distribution, coherent detection, generalized likelihood ratio test (GLRT), sub-Gaussian symmetric alpha-stable (SGS $\alpha$ S) distribution.

## I. INTRODUCTION

COHERENT detection in correlated Gaussian or non-Gaussian clutter background has attached much attention in some areas such as radar, sonar, and so on. The generalized likelihood ratio test (GLRT) [2] is a commonly used criterion for the coherent target detector design. For Gaussian background, some GLRT-based adaptive detectors such as the Kelly's generalized likelihood test (KGLRT), the adaptive matched filter (AMF), and so on have been proposed [1]. Due to more complex environment and more advanced radar equipment, some measured clutter shows non-Gaussian behaviors in practice.

For non-Gaussian background, a commonly used clutter model is the spherically invariant random process (SIRP) model [2], [15]. This model can be interpreted as the speckle component modulated by a texture component. Detectors such as the optimum K detector for gamma texture [3], the GLRT linear-threshold detector (GLRT-LTD) for inverse gamma texture [4], and the GLRT with inverse Gaussian texture detector for inverse Gaussian texture [5] have been proposed.

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Recently, the alpha-stable distribution is used to model the non-Gaussian clutter especially the clutter whose power distribution exhibits heavy tail. Alpha-stable distribution has two significant properties. Firstly, it is the only distribution that satisfies the generalized central limit theorem (GCLT). That means the alpha-stable distribution is the limiting distribution of the sum of the independent and identically distributed (IID) random variables (with finite or infinite variance). Secondly, alpha-stable distribution has the stability property, which means the sum of some independent alpha-stable random variables with the same characteristic exponent parameter  $\alpha$  is still an alpha-stable variable. Besides, the alpha-stable distribution shows an algebraic decaying tail, which means it is suitable for modeling the heavy-tailed clutter. Symmetric alpha-stable (SaS) distribution [6], [7] is a special case of alpha-stable distribution and it has been proved suitable for modeling some measured clutter data. Specifically, SaS distribution is suitable for modeling the real part and the imaginary part of some complex clutter data. In other words, some measured complex clutter data can be modeled as a bivariate isotropic SaS variable. However, the SaS distribution is just suitable for modeling the scalar clutter variable. It is not suitable for modeling the clutter vector in continuous pulses.

Considering radar target coherent detection, the clutter samples in continuous pulses in a range cell are modeled as a random vector. Therefore, a vector form of SaS distribution, the sub-Gaussian symmetric alpha-stable (SGS $\alpha$ S) process, is used to model the clutter vectors [8]. However, due to the problem that the probability density function (PDF) of SGS $\alpha$ S distribution cannot be expressed as a closed form of elementary function except for the cases of multidimensional Gaussian and multidimensional Cauchy distributions, the research on detector design in SGS $\alpha$ S clutter background is very limited.

In this letter, the GLRT-based adaptive detector in SGS $\alpha$ S clutter background (GLRT-SGS $\alpha$ S detector) is formulated. Specifically, the clutter vector is modeled as SGS $\alpha$ S vector. Firstly, the PDF of the SGS $\alpha$ S distribution is derived as a closed-form expression in terms of the Fox's H-function [9]. Then, based on the two-step GLRT criterion, the GLRT-SGS $\alpha$ S detector is formulated and a close-form expression of the test statistics based on the H-function is derived.

The rest of this letter is organized as follows. In Section II, the PDF of SGS $\alpha$ S distribution is derived as a closed form in terms of the H-function. Section II corresponds to the first main contribution of this letter. In Section III, based on the derivation in Section II, the GLRT-based detector in SGS $\alpha$ S clutter background is derived. Section III corresponds to the

second main contribution of this letter. In Section IV, some experimental results are given and discussed. And in Section V, this letter is summarized.

## II. FORMULATION OF SGSaS CLUTTER BASED ON H-FUNCTION

### A. Fox's H-Function

The Fox's H-function has been used in a wide variety of applications. The definition of the Fox's H-function is [9]

$$\begin{aligned} H_{p,q}^{m,n} \left[ z \left| \begin{matrix} (a_1, A_1), \dots, (a_p, A_p) \\ (b_1, B_1), \dots, (b_q, B_q) \end{matrix} \right. \right] \\ = H_{p,q}^{m,n} \left[ z \left| \begin{matrix} (a_1, A_1), \dots, (a_p, A_p) \\ (b_1, B_1), \dots, (b_q, B_q) \end{matrix} \right. \right] \\ = \frac{1}{2\pi i} \int_{\mathcal{L}} \frac{\prod_{j=1}^m \Gamma(b_j + B_j s) \prod_{i=1}^n \Gamma(1 - a_i - A_i s)}{\prod_{i=n+1}^p \Gamma(a_i + A_i s) \prod_{j=m+1}^q \Gamma(1 - b_j - B_j s)} x^{-s} ds \end{aligned} \quad (1)$$

where  $z \neq 0$ ;  $m, n, p$ , and  $q$  are integers with  $0 \leq m \leq q$  and  $0 \leq n \leq p$ ;  $A_i$  and  $B_j$  are positive real numbers;  $a_i$  and  $b_j$  are real or complex numbers;  $\mathcal{L}$  is the integral path, which should be chosen to separate the poles of  $\Gamma(b_j + B_j s)$  from the poles of  $\Gamma(1 - a_i - A_i s)$ .

### B. Formulation of SGSaS Clutter Based on H-Function

The alpha-stable distribution  $S_{\alpha_0}(\sigma, \lambda, \mu)$  is described by its characteristic function (CF) [10]

$$\phi(t) = \begin{cases} \exp \left\{ -\sigma^{\alpha_0} |t|^{\alpha_0} \left[ 1 - i\lambda \tan\left(\frac{\pi\alpha_0}{2}\right) \text{sgn}(t) \right] + i\mu t \right\} & \text{if } \alpha_0 \neq 1 \\ \exp \left\{ -\sigma |t| \left[ 1 + i\frac{2\lambda}{\pi} \text{sgn}(t) \ln(t) \right] + i\mu t \right\} & \text{if } \alpha_0 = 1 \end{cases} \quad (2)$$

where  $\alpha_0 \in (0, 2]$  is the characteristic exponent,  $\sigma > 0$  is the scale exponent,  $\lambda \in [-1, 1]$  represents the skewness, and  $\mu \in \mathbb{R}$  represents the location. When  $\lambda = 0$  and  $\mu = 0$ , the alpha-stable distribution becomes the SaS distribution  $S_{\alpha_0}(\sigma, 0, 0)$ . According to the sub-Gaussian Theory in [10], an SaS variable  $c$  can be written as

$$c = \sqrt{s}g \quad (3)$$

where  $s$  is a positive alpha-stable (PaS) variable [11], and  $g$  is a zero mean Gaussian variable. Then, any complex random vector  $\mathbf{c}$  is an SGSaS vector in  $\mathbb{R}^N$  if it can be written as

$$\mathbf{c} = \sqrt{s}\mathbf{g} \quad (4)$$

where  $\mathbf{g}$  is a zero mean complex Gaussian vector. Here, it is assumed that  $\mathbf{g}$  is an  $N$ -dimensional circular complex Gaussian vector with zero mean and covariance matrix  $\mathbf{R}$ . Therefore, the SGSaS model here is indeed the SIPR model with a PaS distributed texture  $s$ . The CF of the SGSaS vector  $\mathbf{c}$  can be expressed as

$$\phi(\mathbf{t}) = \exp \left[ -\frac{1}{2} (\mathbf{t}^T \mathbf{R} \mathbf{t}) \right]. \quad (5)$$

From (5), it is obvious that the PDF of  $\mathbf{c}$  cannot be expressed as a closed-form expression except for the cases  $\alpha_0 = 1$  (multivariate Cauchy) and  $\alpha_0 = 2$  (multivariate Gaussian).

Next, the closed-form expression of the PDF of SGSaS clutter will be derived in terms of the H-function. According to (4), the PDF of the SGSaS vector can be written as

$$\begin{aligned} f_{\mathbf{c}}(\mathbf{c}) &= \int_0^\infty f_{\mathbf{c}|s}(\mathbf{c}|s) f_s(s) ds \\ &= \int_0^\infty \frac{1}{(\pi s)^N |\mathbf{R}|} \exp \left( -\frac{\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c}}{s} \right) f_s(s) ds \end{aligned} \quad (6)$$

where  $f_{\mathbf{c}|s}(\mathbf{c}|s)$  is the PDF of the random vector  $\mathbf{c}$  conditioned to the PaS variable  $s$  and  $f_s(s)$  is the PDF of  $s$ . According to [11], the PDF  $f_s(s)$  of the PaS variable  $s$  can be expressed as a closed-form expression in terms of the Fox's H-Function

$$f_s(s) = s^{-1} H_{1,1}^{0,1} \left[ \gamma^{-1} s^\alpha \left| \begin{matrix} (1, 1) \\ (1, \alpha) \end{matrix} \right. \right] \quad (7)$$

where the parameter are  $\alpha = \alpha_0/2$  and  $\gamma = \sigma^\alpha / \cos(\pi\alpha/2)$ .

Then, according to (2.9.4), Property 2.2, and Property 2.3 in [9],  $\exp(\cdot)$  in (6) can be expressed as

$$\begin{aligned} \exp \left( -\frac{\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c}}{s} \right) &= H_{0,1}^{1,0} \left[ \frac{\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c}}{s} \left| \begin{matrix} - \\ (0, 1) \end{matrix} \right. \right] \\ &= H_{1,1}^{1,1} \left[ \frac{s}{\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c}} \left| \begin{matrix} (1, 1) \\ (1, 0) \end{matrix} \right. \right]. \end{aligned} \quad (8)$$

Then, substituting (7) and (8) into (6) leads to

$$\begin{aligned} f_{\mathbf{c}}(\mathbf{c}) &= \frac{\pi^{-N}}{|\mathbf{R}|} \int_0^\infty s^{-N-1} H_{1,1}^{0,1} \left[ \gamma^{-1} s^\alpha \left| \begin{matrix} (1, 1) \\ (1, \alpha) \end{matrix} \right. \right] \\ &\quad \times H_{1,1}^{1,1} \left[ \frac{s}{\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c}} \left| \begin{matrix} (1, 1) \\ (1, 0) \end{matrix} \right. \right] ds. \end{aligned} \quad (9)$$

Then, according to Theorem 2.9 in [9],  $f_{\mathbf{c}}(\mathbf{c})$  can be finally expressed as a closed-form expression

$$\begin{aligned} f_{\mathbf{c}}(\mathbf{c}) &= \frac{\pi^{-N}}{|\mathbf{R}|} (\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c})^{-N} H_{1,2}^{1,1} \\ &\quad \times \left[ \gamma^{-1} (\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c})^\alpha \left| \begin{matrix} (1, 1) \\ (N, \alpha)(1, \alpha) \end{matrix} \right. \right] \\ &\equiv \frac{\pi^{-N}}{|\mathbf{R}|} (\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c})^{-N} H \left[ \gamma^{-1} (\mathbf{c}^H \mathbf{R}^{-1} \mathbf{c})^\alpha \right]. \end{aligned} \quad (10)$$

### C. Fitting Measured Data With SGSaS Distribution

Here, we test the performances of SGSaS distribution to fit the measured data compared with some other common distributions such as the K, Weibull, Log-normal, and generalized Pareto (GP) distributions. Here, datasets 1 and 2 are selected from the well-known IPIX dataset and dataset 3 is selected from the sea-detecting radar data-sharing program dataset [14]. These three datasets are, respectively, collected at 22:40:24, February 4, 1998, from HV polarization channel; 18:47:33, February 5, 1998, from HH polarization channel; and 11:24:46, October 12, 2019, from HH polarization channel. Fig. 1 shows the histogram of measured sea clutter and the PDFs of five distributions. From Fig. 1, we can see that the SGSaS distribution performs better in fitting the measured data tails than other four distributions.

Then, the well-known Kolmogorov–Smirnov (KS) statistic [5] is also adopted to evaluate fitting results. Table I

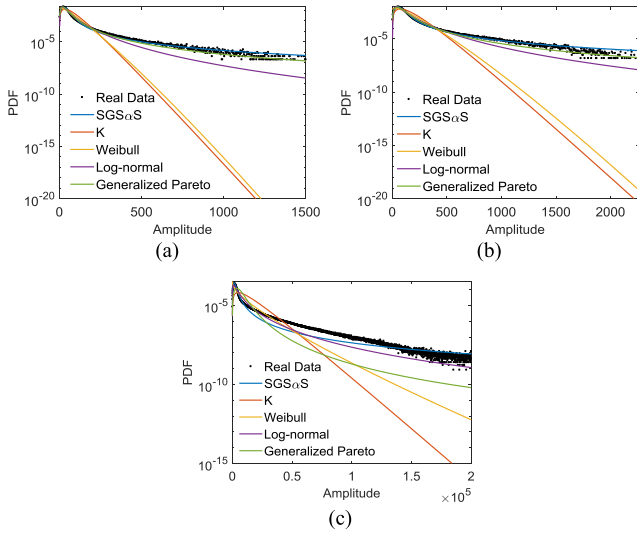


Fig. 1. Histogram of measured sea clutter and the PDFs of four commonly-used distributions. (a) Dataset 1. (b) Dataset 2. (c) Dataset 3.

TABLE I

KS STATISTICS OF DIFFERENT DISTRIBUTIONS FOR FITTING MEASURED CLUTTER DATA

| Dataset | SGS $\alpha$ S | K      | Weibull | Log-Normal | GP     |
|---------|----------------|--------|---------|------------|--------|
| 1       | 0.0182         | 0.2566 | 0.1093  | 0.0506     | 0.0414 |
| 2       | 0.0213         | 0.2377 | 0.1020  | 0.0462     | 0.0375 |
| 3       | 0.0760         | 0.4900 | 0.1517  | 0.0941     | 0.3375 |

gives the KS statistics of five distributions for fitting the measured clutter data. From Table I, we can see that the SGS $\alpha$ S distribution has the best fitting performance among the five distributions.

### III. GLRT-BASED DETECTOR IN SGS $\alpha$ S CLUTTER

Here, the clutter-dominant environment is considered and then the thermal noise is ignored. The detection problem can be seen as a binary hypothesis testing problem. The received signal  $\mathbf{z}$  is an  $N \times 1$  column vector. Under the condition that target is absent ( $H_0$ ), the received signal vector  $\mathbf{z}$  consists of clutter vector  $\mathbf{c}$ ; under the condition that target is present ( $H_1$ ),  $\mathbf{z}$  consists of clutter vector  $\mathbf{c}$  and target signal vector  $\mathbf{s}$

$$\begin{cases} H_0 : \mathbf{z} = \mathbf{c} \\ H_1 : \mathbf{z} = \mathbf{s} + \mathbf{c}. \end{cases} \quad (11)$$

The target signal vector  $\mathbf{s}$  is modeled as  $\mathbf{s} = \beta \mathbf{p}$ , where  $\beta$  is the complex amplitude of the target and  $\mathbf{p}$  is the steering vector. In this letter, it is assumed that  $\mathbf{p}$  is known and  $\beta$  is an unknown but deterministic parameter. The covariance matrix  $\mathbf{R}$  is also assumed unknown. Next, the two-step GLRT detector in SGS $\alpha$ S clutter background will be designed. Note that the proposed detector here is also the detector for the SIRP model with a PaS texture.

#### A. Step 1

In Step 1, it is assumed that  $\beta$  is deterministic but unknown and  $R$  is known. The likelihood ratio test can be written as

$$\Lambda(x) = \frac{\max_{\beta} f_{\mathbf{x}|H_1}(\mathbf{z}|\beta, H_1)}{f_{\mathbf{x}|H_0}(\mathbf{z}|H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} T \quad (12)$$

where  $\Lambda(x)$  is the detection statistic,  $f_{\mathbf{x}|H_0}(\mathbf{z}|H_0)$  is the PDF of  $\mathbf{z}$  under  $H_0$ ,  $f_{\mathbf{x}|H_1}(\mathbf{z}|\beta, H_1)$  is the PDF of  $\mathbf{z}$  under  $H_1$ , and  $T$  is the threshold. According to (9),  $f_{\mathbf{x}|H_0}(\mathbf{z}|H_0)$  and  $f_{\mathbf{x}|H_1}(\mathbf{z}|\beta, H_1)$  can be written as

$$\begin{aligned} f_{\mathbf{x}|H_0}(\mathbf{z}|H_0) &= f_{\mathbf{c}}(\mathbf{z}|H_0) \\ &= \frac{\pi^{-N}}{|\mathbf{R}|} (\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z})^{-N} H \left[ \gamma^{-1} (\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z})^{\alpha} \right] \end{aligned} \quad (13)$$

$$\begin{aligned} f_{\mathbf{x}|H_1}(\mathbf{z}|\beta, H_1) &= f_{\mathbf{c}}(\mathbf{z} - \beta \mathbf{p}) \\ &= \frac{\pi^{-N}}{|\mathbf{R}|} [(\mathbf{z} - \beta \mathbf{p})^H \mathbf{R}^{-1} (\mathbf{z} - \beta \mathbf{p})]^{-N} \\ &\quad \times H \left[ \gamma^{-1} ((\mathbf{z} - \beta \mathbf{p})^H \mathbf{R}^{-1} (\mathbf{z} - \beta \mathbf{p}))^{\alpha} \right]. \end{aligned} \quad (14)$$

Substituting (13) and (14) into (12), we get

$$\frac{[(\mathbf{z} - \beta \mathbf{p})^H \mathbf{R}^{-1} (\mathbf{z} - \beta \mathbf{p})]^{-N} H \left[ \gamma^{-1} ((\mathbf{z} - \beta \mathbf{p})^H \mathbf{R}^{-1} (\mathbf{z} - \beta \mathbf{p}))^{\alpha} \right]}{(\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z})^{-N} H \left[ \gamma^{-1} (\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z})^{\alpha} \right]} \underset{H_0}{\overset{H_1}{\gtrless}} T. \quad (15)$$

#### B. Step 2

In the second step, the maximum likelihood estimate  $\hat{\beta}$  is obtained to replace  $\beta$  in (15) and the fixed point (FP) estimate  $\hat{\mathbf{R}}_{\text{FP}}$  [12] is obtained to replace  $\mathbf{R}$  in (15). The maximum likelihood estimate  $\hat{\beta}$  is known as

$$\hat{\beta} = \frac{\mathbf{p}^H \mathbf{R}^{-1} \mathbf{z}}{\mathbf{p}^H \mathbf{R}^{-1} \mathbf{p}}. \quad (16)$$

The FP estimate  $\hat{\mathbf{R}}_{\text{FP}}$  of covariance matrix  $\mathbf{R}$  is

$$\hat{\mathbf{R}}_{\text{FP}} = \frac{N}{K} \sum_{k=1}^K \frac{\mathbf{z}_k \mathbf{z}_k^H}{\mathbf{z}_k^H \hat{\mathbf{R}}_{\text{FP}}^{-1} \mathbf{z}_k} \quad (17)$$

where  $K$  is the number of secondary vector data which are used to estimate the covariance matrix.

$\hat{\mathbf{R}}_{\text{FP}}$  can be solved by the following iterative process:

$$\hat{\mathbf{R}}_{\text{FP}}^{(j+1)} = \frac{N}{K} \sum_{k=1}^K \frac{\mathbf{z}_k \mathbf{z}_k^H}{\mathbf{z}_k^H (\hat{\mathbf{R}}_{\text{FP}}^{(j)})^{-1} \mathbf{z}_k} \quad (18)$$

with

$$\hat{\mathbf{R}}_{\text{FP}}^{(j+1)} = \frac{N}{\text{Tr}(\hat{\mathbf{R}}_{\text{FP}}^{(j+1)})} \hat{\mathbf{R}}_{\text{FP}}^{(j+1)} \quad (19)$$

and initial value

$$\hat{\mathbf{R}}_{\text{FP}}^{(0)} = \frac{N}{K} \sum_{k=1}^K \frac{\mathbf{z}_k \mathbf{z}_k^H}{\mathbf{z}_k^H \mathbf{z}_k}. \quad (20)$$

Substituting  $\hat{\mathbf{R}}_{\text{FP}}$  into (15), we get

$$\begin{aligned} &\left[ 1 - \frac{|\mathbf{p}^H \hat{\mathbf{R}}_{\text{FP}}^{-1} \mathbf{z}|^2}{(\mathbf{p}^H \hat{\mathbf{R}}_{\text{FP}}^{-1} \mathbf{p})(\mathbf{z}^H \hat{\mathbf{R}}_{\text{FP}}^{-1} \mathbf{z})} \right]^{-N} \\ &\quad \times \frac{H \left[ \gamma^{-1} \left( \mathbf{z}^H \hat{\mathbf{R}}_{\text{FP}}^{-1} \mathbf{z} - \frac{|\mathbf{p}^H \hat{\mathbf{R}}_{\text{FP}}^{-1} \mathbf{z}|^2}{\mathbf{p}^H \hat{\mathbf{R}}_{\text{FP}}^{-1} \mathbf{p}} \right)^{\alpha} \right]}{H \left[ \gamma^{-1} (\mathbf{z}^H \hat{\mathbf{R}}_{\text{FP}}^{-1} \mathbf{z})^{\alpha} \right]} \underset{H_0}{\overset{H_1}{\gtrless}} T. \end{aligned} \quad (21)$$

### C. CFAR Property Against Covariance Matrix $\mathbf{R}$

Here, the CFAR property of this detector will be analyzed. There is always a unitary matrix  $\mathbf{U}$  such that

$$\mathbf{e}_1 = \frac{\mathbf{U}\mathbf{R}^{-1/2}\mathbf{p}}{(\mathbf{p}^H\mathbf{R}^{-1}\mathbf{p})^{1/2}} \quad (22)$$

where  $\mathbf{e}_1$  is an  $N$ -dimensional unit vector:  $\mathbf{e}_1 = [1, 0, \dots, 0]^T$ , and  $(\cdot)^T$  means the transpose. Then, let

$$\bar{\mathbf{z}} = \mathbf{U}\mathbf{R}^{-1/2}\mathbf{z} \quad (23)$$

and

$$\bar{\mathbf{R}} = \mathbf{U}\mathbf{R}^{-1/2}\hat{\mathbf{R}}_{\text{FP}}^{-1}\mathbf{R}^{-1/2}\mathbf{U}^H. \quad (24)$$

Then, the two-step GLRT-based detector in (21) becomes

$$\left[ 1 - \frac{|\mathbf{e}_1^H \bar{\mathbf{R}}^{-1} \bar{\mathbf{z}}|^2}{(\mathbf{e}_1^H \bar{\mathbf{R}}^{-1} \mathbf{e}_1)(\bar{\mathbf{z}}^H \bar{\mathbf{R}}^{-1} \bar{\mathbf{z}})} \right] \times \frac{H \left[ \gamma^{-1} \left( \bar{\mathbf{z}}^H \bar{\mathbf{R}}^{-1} \bar{\mathbf{z}} - \frac{|\mathbf{e}_1^H \bar{\mathbf{R}}^{-1} \bar{\mathbf{z}}|^2}{\mathbf{e}_1^H \bar{\mathbf{R}}^{-1} \mathbf{e}_1} \right)^\alpha \right]}{H \left[ \gamma^{-1} (\bar{\mathbf{z}}^H \bar{\mathbf{R}}^{-1} \bar{\mathbf{z}})^\alpha \right]} \underset{H_0}{\overset{H_1}{\geq}} T. \quad (25)$$

Under  $H_0$ ,  $\bar{\mathbf{z}}$  in (25) can be expressed by  $\bar{\mathbf{z}} = \sqrt{s}\bar{\mathbf{g}}$  with  $\bar{\mathbf{g}} = \mathbf{U}\mathbf{R}^{-1/2}\mathbf{g}$ . Here,  $\bar{\mathbf{g}}$  is a circularly symmetric, complex Gaussian vector. In [12],  $\bar{\mathbf{R}}$  has been proved independent on the covariance matrix  $\mathbf{R}$ . Then, we can see that all the parameters in (25) are independent of the covariance matrix  $\mathbf{R}$ . Therefore, the two-step GLRT detector derived in this letter is CFAR against  $\mathbf{R}$ . Notably, the same as the adaptive normalized matched filter (ANMF) with  $\hat{\mathbf{R}}_{\text{FP}}$  in [18], the proposed detector is not CFAR against  $s$ . In fact, the same as the ANMF, the proposed detector is also a uniformly most powerful invariant (UMPI) detector [19], [20].

## IV. EXPERIMENTAL RESULTS

### A. Results on Simulated Data

Firstly, the thresholds are obtained through  $10^8$  Monte Carlo simulations according to the set probability of false alarm (Pfa). Then, clutter and target samples are simulated and different detectors are applied for the simulated data. Note that the H-function is not available in standard packages of most software. In this letter, the numerical method to calculate the H-function in [13] is adopted.

The parameters are set as follows: the length of received vector  $N = 16$ , the number of secondary vector  $K = 32$ , the parameter in PaS distribution  $\gamma = 1$ , and the steering vector  $\mathbf{p} = 1/\sqrt{N}[1, \exp(j2\pi f_d), \dots, \exp(j2\pi(N-1)f_d)]^T$  with  $f_d = 0.3$ . The clutter covariance matrix is set as

$$[\mathbf{R}]_{ij} = \rho^{|i-j|}, \quad 1 \leq i, j \leq N \quad (26)$$

where  $\rho$  is the one-lag correlation coefficient and set to 0.9.

The variance of alpha-stable distribution is infinite. Therefore, the traditional signal-to-clutter ratio (SCR) is no longer effective. Here, a generalized signal-to-clutter ratio (GSCR) [11] is adopted

$$\text{GSCR} = 10 \log \left[ \left( \frac{\beta}{\gamma^{1/\alpha}} \right)^\alpha \right]. \quad (27)$$

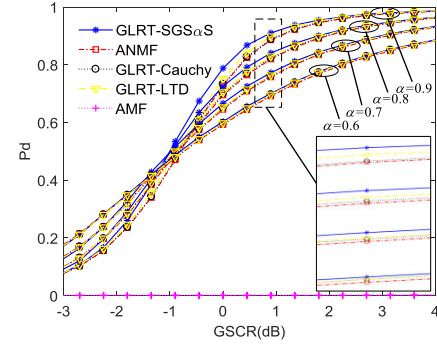


Fig. 2. Pd versus GSCR comparison for various  $\alpha$  values and  $\text{Pfa} = 10^{-3}$ .

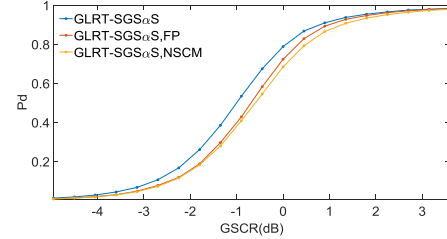


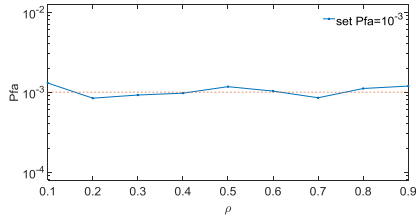
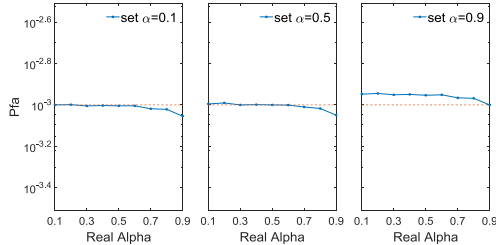
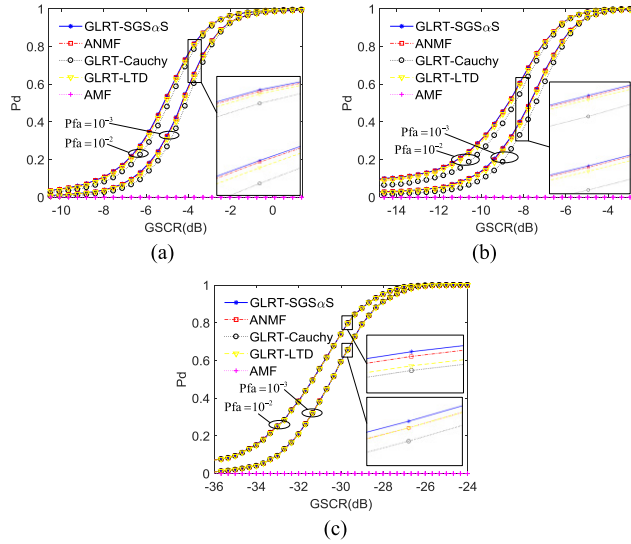
Fig. 3. Pfa versus GSCR comparison for different covariance matrix estimation methods with  $\text{Pfa} = 10^{-3}$ .

Firstly, the detection performance of the proposed GLRT-SGSaS detector is evaluated and compared with some other detectors. The comparison methods are the AMF [1], the ANMF [16]–[19], the GLRT method for Cauchy clutter background (GLRT-Cauchy) [8], and the GLRT-LTD [4]. Among the comparison methods, the AMF and GLRT-Cauchy are suitable for the Gaussian and Cauchy clutter, respectively, which are the two special cases of SGSaS clutter. The GLRT-LTD and ANMF are suitable for the detection in heavy-tailed clutter background. Fig. 2 shows the probability of detection (Pd) versus GSCR comparison for  $\alpha = 0.9, 0.8, 0.7$ , and  $0.6$ , respectively, with set  $\text{Pfa} = 10^{-3}$ . We can see that the proposed GLRT-SGSaS detector has the best performance compared with the comparison methods for different values of  $\alpha$ . And with the increase of  $\alpha$ , the performance gap between the proposed method and the comparison methods becomes larger. In addition, the AMF method designed for Gaussian background is no longer effective in SGSaS background because SGSaS distribution seriously deviates from the Gaussian distribution.

Then, the performance FP covariance matrix estimation method is evaluated. Fig. 3 gives the Pd versus GSCR using the covariance matrix  $\mathbf{R}$ , the FP estimate  $\hat{\mathbf{R}}_{\text{FP}}$ , and the normalized sample covariance matrix (NSCM) estimate  $\hat{\mathbf{R}}_{\text{NSCM}}$  [2] with set  $\text{Pfa} = 10^{-3}$ . From Fig. 3, it is obvious that the GLRT-SGSaS with the ideal covariance matrix  $\mathbf{R}$  has the best detection performance. And the GLRT-SGSaS with FP estimate  $\hat{\mathbf{R}}_{\text{FP}}$  proposed in this letter has a better performance than the GLRT-SGSaS with NSCM estimate  $\hat{\mathbf{R}}_{\text{NSCM}}$ .

Fig. 4 gives the Pfa of the proposed GLRT-SGSaS with FP estimate  $\hat{\mathbf{R}}_{\text{FP}}$  versus the one-lag correlation coefficient  $\rho$  in covariance matrix  $\mathbf{R}$ . From Fig. 4, we can see that the Pfa hardly changes with  $\rho$ , which indicates the proposed method is independent of  $\rho$ . In other words, the proposed method is independent of  $\mathbf{R}$ .



Fig. 4. CFAR property verification with set  $P_{fa} = 10^{-3}$ .Fig. 5. Mismatch analysis results with set  $P_{fa} = 10^{-3}$ .Fig. 6.  $P_d$  versus GSCR on the IPIX data. (a) Dataset 1. (b) Dataset 2. (c) Dataset 3.

To analyze the robustness of the proposed detector when the set  $\alpha$  for proposed detector and the real  $\alpha$  of SGS $\alpha$ S clutter mismatch, we set a fixed  $\alpha$  for the proposed detector and test the performance change with the real  $\alpha$ . Fig. 5 gives some mismatch analysis results with set  $P_{fa} = 10^{-3}$ . From Fig. 5, we can see that the real  $P_{fa}$  is little higher than the ideal  $P_{fa}$  when the real value of  $\alpha$  is smaller than the set value of  $\alpha$ , and otherwise the opposite. However, the gap between the real  $P_{fa}$  and the set  $P_{fa}$  is small. Therefore, we can conclude that the proposed detector has some robustness with respect to the parameter  $\alpha$  to some extent.

### B. Results on Measured Data

Here, the experimental results on three measured datasets used in Section II-C are presented. The target data are simulated and added to the three datasets. Fig. 6 gives the  $P_d$  of the proposed method and 4 other comparison methods versus GSCR on the three datasets. From Fig. 6, we can see that the proposed detector has the best detection performance for all

the three datasets. These results also verify the effectiveness of the proposed GLRT-SGS $\alpha$ S detector in this letter.

## V. CONCLUSION

In this letter, a GLRT-based detector in SGS $\alpha$ S background, the GLRT-SGS $\alpha$ S detector, is proposed. The closed-form PDF expression of SGS $\alpha$ S distribution is firstly formulated in terms of the Fox's H-function. Then, the GLRT-SGS $\alpha$ S detector is formulated in terms of the Fox's H-function based on the two-step GLRT criterion. The correctness of the derivation process and the effectiveness of the proposed detector are verified by experiments on simulated and measured data.

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